

Sno	Authors	Title of the Paper	Year of Published	Dataset Used	Methodology Used	Limitation of the Paper
1	G. K. et al.	Random Forest Modeling for Network Intrusion Detection System	2016	NSL-KDD	Random Forest	Uses an older benchmark dataset
2	Arif Jamal Malik, Waseem Shahzad, Farrukh Aslam Khan	Network Intrusion Detection Using Hybrid Binary PSO and Random Forests Algorithm	2015	KDD99	Binary PSO + Random Forest	Uses an old dataset
3	Iftikhar Ahmad, Mohammad Basher, Mohammad Javed Iqbal, Aneel Rahim	Performance Comparison of Support Vector Machine, Random Forest, and Extreme Learning Machine for Intrusion Detection	2018	NSL-KDD	SVM, Random Forest, Extreme Learning Machine	Performance depends on the selected dataset and features
4	Yang Jia, Meng Wang, Yagang Wang	Network Intrusion Detection Algorithm Based on Deep Neural Network	2019	KDD99, NSL-KDD	Deep Neural Network	Mainly uses older benchmark datasets
5	Chuanlong Yin, Yuefei Zhu, Jinlong Fei, Xinzhen He	A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks	2017	KDD99	Recurrent Neural Network	Requires more computational resources
6	Khaled Alrawashdeh, Carla Purdy	Toward an Online Anomaly Intrusion Detection System Based on Deep Learning	2016	KDD99	Deep Belief Network, Restricted Boltzmann Machine	Uses an older dataset
7	Yang Yu, Jun Long, Zhiping Cai	Network Intrusion Detection through Stacking Dilated Convolutional Autoencoders	2017	KDD99	Deep Learning, Convolutional Autoencoder	Deep model is complex and time-consuming to train

8	Yong Zhang, Xu Chen, Lei Jin, Xiaojuan Wang, Da Guo	Network Intrusion Detection: Based on Deep Hierarchical Network and Original Flow Data	2019	Network Flow Data	Deep Hierarchical Network	Requires high computational resources
9	Sunanda Gamage, Jagath Samarabandu	Deep Learning Methods in Network Intrusion Detection: A Survey and an Objective Comparison	2020	KDD99, NSL- KDD, CIC- IDS2017, CIC- IDS2018	Feedforward Neural Network, Autoencoder, DBN, LSTM	Comparison mainly depends on benchmark datasets
10	Hafiza Anisa Ahmed, Anum Hameed, Narmeen Zakaria Bawany	Network Intrusion Detection Using Oversampling Technique and Machine Learning Algorithms	2022	UNSW-NB15	Random Forest, Decision Tree, Logistic Regression, KNN, ANN, SMOTE	Class imbalance remains a challenge
11	Sharuka Promodya Thirimanne et al.	Deep Neural Network Based Real-Time Intrusion Detection System	2022	NSL-KDD	Deep Neural Network, Real-Time Feature Extraction	Real-time implementation is complex
12	Nour Moustafa, Jill Slay	UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems	2015	UNSW-NB15	Dataset Generation and Analysis	Benchmark dataset may not represent every real- world network
13	Nour Moustafa, Jill Slay	The Evaluation of Network Anomaly Detection Systems: Statistical Analysis of the UNSW-NB15 Data Set and Comparison with KDD99	2016	UNSW-NB15, KDD99	Statistical Analysis	Dataset comparison rather than a complete IDS
14	Markus Ring, Sarah Wunderlich, Deniz Scheuring, Dieter Landes, Andreas Hotho	A Survey of Network-Based Intrusion Detection Data Sets	2019	Multiple IDS Datasets	Dataset Survey and Comparison	Does not develop a new detection model
15	Geeta Kocher, Gulshan Kumar	Machine Learning and Deep Learning Methods for Intrusion Detection Systems: Recent Developments and Challenges	2021	Multiple Datasets	Machine Learning, Deep Learning	Review paper; does not propose one new model

16	M. A. Ferrag et al.	Deep Learning for Cyber Security Intrusion Detection: Approaches, Datasets, and Comparative Study	2020	CSE-CIC-IDS2018, Bot-IoT	DNN, RNN, CNN, Random Forest	Different datasets make direct comparison difficult
17	Yifan Tang, Lize Gu, Leiting Wang	Deep Stacking Network for Intrusion Detection	2022	NSL-KDD	Decision Tree, KNN, DNN, Random Forest, Deep Stacking	Model becomes more complex
18	Neha Sharma, Narendra Singh Yadav, Saurabh Sharma	Classification of UNSW-NB15 Dataset Using Exploratory Data Analysis Using Ensemble Learning	2021	UNSW-NB15	Random Forest, Extra Trees, AdaBoost, XGBoost	Focuses mainly on ensemble methods
19	Soulaiman Moualla et al.	Improving the Performance of Machine Learning-Based Network Intrusion Detection Systems on the UNSW-NB15 Dataset	2021	UNSW-NB15	Feature Selection, Machine Learning, SMOTE	Class imbalance affects performance
20	Aklil Zenebe Kiflay, Athanasios Tsokanos, Mahmood Fazlali, Raimund Kirner	Network Intrusion Detection Leveraging Multimodal Features	2024	UNSW-NB15	Two Random Forest Models, Soft Voting	Practical online implementation was not evaluated
21	Humera Ghani, Bal Virdee, Shahram Salekzamankhani	A Deep Learning Approach for Network Intrusion Detection Using a Small Features Vector	2023	UNSW-NB15, NSL-KDD	Feedforward Neural Network	Uses only a small selected feature set
22	Mohammad A. Alsharaiah et al.	An Innovative Network Intrusion Detection System: Hierarchical Deep Learning Model Based on UNSW-NB15 Dataset	2024	UNSW-NB15	LSTM, Attention Mechanism	Deep learning requires more resources

23	Sow Thierno Hamidou, Adda Mehdi	Enhancing IDS Performance Through a Comparative Analysis of Random Forest, XGBoost, and Deep Neural Networks	2025	NSL-KDD	Random Forest, XGBoost, DNN, SMOTE	Mainly evaluated using NSL-KDD
24	Muhammad Farhan et al.	Network-Based Intrusion Detection Using Deep Learning Technique	2025	UNSW-NB15	Deep Learning, Extra Trees	Generalization to other datasets needs further study
25	Vipin Jain, Kirti Singh, Suneet Joshi, Lokesh Malviya et al.	Optimizing Intrusion Detection Using a Hybrid CNN-LSTM Deep Learning Framework on the NSL-KDD Dataset	2026	NSL-KDD	CNN, LSTM	Hybrid deep learning models can be computationally expensive